535/1
Physics
Paper 1 (Theory)
2026

BUGANDA EXAMINATION COUNCIL

Uganda Certificate of Education

Physics

Paper 1 (Theory)

2 hours 30 minutes

INSTRUCTIONS TO CANDIDATES

- This examination paper consists of two sections A and B.
- It has 7 examination items
- Section A has three compulsory items
- Section B has two parts; I and II. Respond to only 1 item from each part.
- Respond to 5 items in all
- Any additional item(s) responded to will not be scored.
- All responses must be written in the answer booklet(s) provided.
- Responses to each item should start on a fresh page.
- Indicate the item number you are responding to, clearly at the top of the page.
- Graph paper is provided
- Mathematical table and silent non-programmable calculators may be used.

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SECTION A

Respond to all items in this section

Item 1

A house hold resides close to a church where night prayer sessions are regularly held. On one occasion, a member of the family noticed that the singing and praise during the night seemed much louder than what is usually experienced in day time services. This led him to suspect that something unusual might be happening, so he chose to go and observe the situation at the church, accompanied by his daughter who was dressed in a red dress. While there, he became even more curious about how the girl's dress appeared when it was lit by a source producing red and yellow lights used for signaling to the distant worshippers due to their frequencies.

Hint:

- Speed of light in air = $3.0 \times 10^{8} ms^{-1}$
- Wave length of red light = $7.5 \times 10^{-7} m$
- Wave length of yellow = $5.8 \times 10^{-7} m$.
- Light of a lower frequency travels a longer distance.

Task

As a student of physics, help;

(a) explain the family member's observations using scientific principles.
(b) the daughter complete her assignment of identifying which colour can easily be recognized by distant worshippers.

Item 2

While touring a hospital, a visitor was accidentally splashed by a liquid from a container. He was informed that the substance was radioactive, with an activity of 450 counts per second and a half-life of two days. As a precaution, he was required to remain isolated in the hospital until the count rate decreased to 6.25 counts per second.

Concerned by the incident, his relatives asked the hospital management to improve the methods used to store such materials.

Hint:

Background radiation is 50 counts per second.

Task:

As a student of Physics,

(a) Assist the patient in calculating the duration he will need to remain in the hospital.

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Task:

As a Physics student, help the man to;

(a) Understand why the braking system of his car is hot.

(b) Understand why the car was unstable on starting to move and explain how it can be addressed.

(c) Know the time taken for the car to come to rest.

(d) Determine the amount of heat absorbed by the material of the system and state any assumption made.

Item 5

A water pump is used to fill a storage tank by lifting water from an underground well for use in the kitchen at school. To reduce power consumption and limit operating time, the pump runs at a constant power of 1800W. It is capable of raising $0.200m^3$ of water every minute through a vertical height of 8m.

In the kitchen, the cook often notices that water pumped in the morning has a temperature higher than expected about 32°C when measured with a thermometer.

The pump is considered efficient if the ratio of useful work output to energy input exceeds 0.75,

for preparing morning tea, the cook heats 8.5 litres of water during which 600g of the water is converted into steam.

Hint:

- Density of water = $1000kgm^{-3}$
- Specific heat capacity of water = $4200Jkg^{-1}k^{-1}$
- Specific latent heat of vaporization of water = $2.56 \times 10^6 Jkg^{-1}$
- Acceleration due to gravity, g = $10ms^{-2}$

Task

As a physics student help the cooks to

(a) Understand the rate at which the pump works in raising the water

(b) Confirm whether the pump is efficient.

(c) Explain why water pumped in the morning has such temperature

(d) Determine how much heat is necessary to turn the lost volume of water into vapour and heat required to raise that evaporated water from an initial temperature.

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Part II

Respond to one item from this part

Item 6

In a certain municipality in Uganda, a growing local packaging company has recently invested in an imported machine to increase the production of plastic bags and containers. The purpose of acquiring this machine was to improve output and enable the company to compete with large industries.

However, soon after installation, the technical team identified a significant issue. The machine requires a supply voltage of 150V, whereas the available main supply from the national grid is 240V. To prevent damage to the machine a chief technician suggested modifying a power regulation set up by incorporating a transformer with a primary coil of 120 turns. The team is uncertain about the required number of turns in the secondary coil to obtain the correct output voltage.

The company intends to operate the machine rated 960W for four hours each working day. A weekly electricity budget of 10,000 Uganda shillings has been set based on a tariff of 1000 shillings per unit (Kwh).

Because of frequent power outages in the area, the company owner is also considering installing a backup energy storage system using a battery to ensure continuous operation during blackouts. He is seeking practical guidance on how to properly maintain the battery and extend its service life.

To address these concerns, the company manager gathers the technical staff and some employees during a break and asks for clear practical explanations so that the system can be implemented safely and economically.

Hint: The Company operates from Monday to Friday.

Task:

As a physics learner;

(a)(i) Help the technical team to determine the correct number of turns required in the other coil so that the machine works safely and identify the machine.

(ii) With the aid of a diagram, describe the principle of operation of the above machine.

(b) Determine whether the company's weekly budget is sufficient to cover the electricity cost of running the machine.

(c) Advise the owner on how to care for and maintain the rechargeable battery backup system to ensure it remains functional, reliable and long lasting.

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Item 7

In a busy wood working shop, a skilled carpenter spends long hours making custom furniture such as beds, tables and chairs because demand is high. To work faster and more accurately, he uses a reliable electric drill rated at 600W and 240V with variable speed. The drill helps him drive screws smoothly into hard woods like mahogany and teak, allowing him to complete many customer orders before end of year celebrations.

One afternoon, while putting together a large wardrobe, the drill suddenly gives off a burning smell, produces a few sparks and then stops rotating completely. His work comes to a halt, putting his deadlines at risk. Feeling frustrated but determined, he takes the drill to a nearby electric technician in a local trading center.

After examining it, the technician explains that the armature coil inside the drill has burnt out. This happened because the drill was used continuously under heavy load without breaks, accumulated a lot of sawdust, was forced to drive very hard screws or experienced unstable voltage from the power supply. The heat damaged the insulation around the wire leading to short circuits within the motor.

However, the technician reassures him that the drill can be repaired to work more efficiently, last longer and produce less heat during operation.

Task:
As a learner of Physics;

(a) Describe the structure and mode of operation of the drill.
(b) Determine the resistance of the drill
(c) Explain two factors that affect the efficiency of the drill and suggest ways of minimizing them.
(d) Advise the carpenter on ways of preventing another burn out in the future.

END

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(b) Explain to the relatives the potential hazards linked to exposure to such liquids.
(c) Guide the hospital personnel on the proper procedures for managing and handling such materials safely.

Item 3

A certain tourist was surprised by a number of observations he made while on a trip in another country.

- The appearance of the moon kept on changing periodically.
- A live program aired on television at night from his home country showed it was day time back home.
- When he opened the car door, a LED light switched itself on. He was surprised when the light automatically switched itself off when the door was locked. He wondered why such lights were used instead of normal bulbs.

The tourist couldn't make sense of the observation made and came to you for help.

Task

Use your knowledge of Physics to help the tourist understand the observation he made and help him understand why such lights were preferred.

SECTION B

Part 1

Respond to one item from this part.

Item 4

A man plans to take his family out for a Christmas Day celebration. He places some luggage in the boot of his car and secures additional bags on the roof rack he then sets off with his 2000kg car from rest and accelerates until it reaches a speed of 30m/s.

After noticing that the car is not stable on the road, he applies the brakes, bringing the vehicle to rest over a distance of 30m. Upon inspecting the car, he finds that the braking system which has a total mass of 30kg, has become extremely hot. This observation leaves him puzzled.

Hint:

- The material of the braking system has specific heat capacity of $1200Jkg^{-1}k^{-1}$
- The temperature increases by 5°C.

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